Lab

Polymorphism

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Lab: Inheritance — towards polymorphism

Objective

The primary objective for this lab is to enable you to derive new types and to override functionality.

Overview

Read the instructions below and critically evaluate each code sample.

Part 1 — Racing cars

Scenario

You are going to work with a project which consists of a `Car` class, a `RacingCar` class, and a `Program (test)` class that fills an **ArrayList**`<Car>` with Car and RacingCar objects.

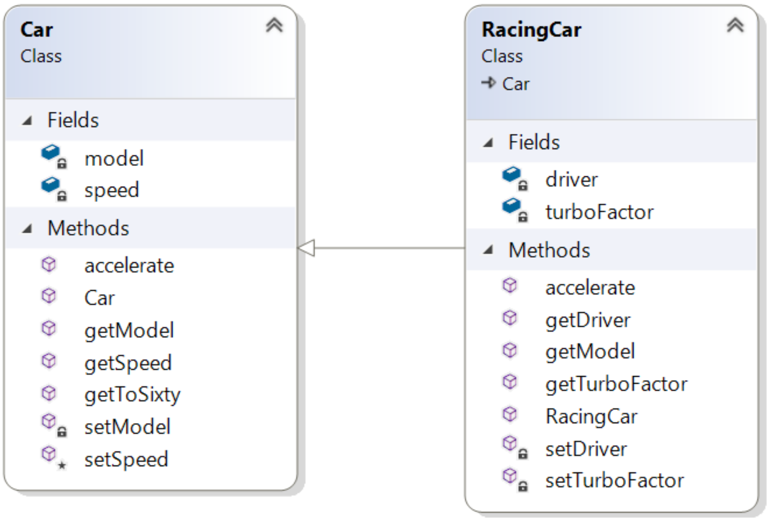
The `Program` class will then process each `Car`, setting their initial speeds to 60MPH (this is done using a blatant cheat that normal cars don't usually possess – a `getToSixty()` method). The test will then continue by making each '`Car`' `accelerate` for two seconds before writing their `model` and `speed` to the `Console`.

If the car is a `RacingCar` we will need to add some additional code to the `Program` class so that a little extra information is written out.

Step-by-step

1. Open the **labs** project.
2. Add a class called **Program** with a **main()** method in a new package called **lab8**.
3. Add a class called **Car** to the **lab8** package.
4. Add another class called **RacingCar** which extends Car.
5. The **Car** and **RacingCar** classes will have the fields and methods as seen in the class diagram below.
6. You will also need to add constructors for the two classes.

Please study the diagram below before writing code. Please also note that any method shown below with the same name as the class is the constructor for that class.



7. Add code to the `Car` class.

   Create getters and setters for the **model** and **speed** fields.

   The **getToSixty()** method just sets the **speed** field to 60.

   The accelerate method should look like **void accelerate**(**int** seconds)

   and increase the speed by 5 * seconds.

8. Create a suitable constructor for the Car class.

9. Add extra code to the **RacingCar** class.

   Add a **String** driver field with getters and setters.

   Add an **int** turboFactor with getters and setters.

10. You'll have to create a suitable constructor for the **RacingCar** class.

11. The **accelerate()** method will invoke the base class (**Car**) accelerate method and then multiply the **speed** by the **turboFactor**.

    Tip: Use the **super.accelerate()** method.

12. The **main()** method should create an array of Cars comprising of a few cars and racing cars.

13. The main() method should then pass the cars array to a method called **`processCars`**.

    The `processCars` method currently:

    - Gets each '`Car`' up to 60MPH as a start point
    - Then accelerates each `Car` for 2 seconds
    - And displays details of each car.

- You'll also display the driver's name but only if a `Car` in the array is a `RacingCar`.

- You'll need to inspect the type of each car in the array. If the car is an instance of RacingCar, cast it to the RacingCar type to access and retrieve the driver's name.
- **Tip:** Use **instanceof** like: if (**c instanceof** *RacingCar*) {....} where '**c**' is a car element in the array.

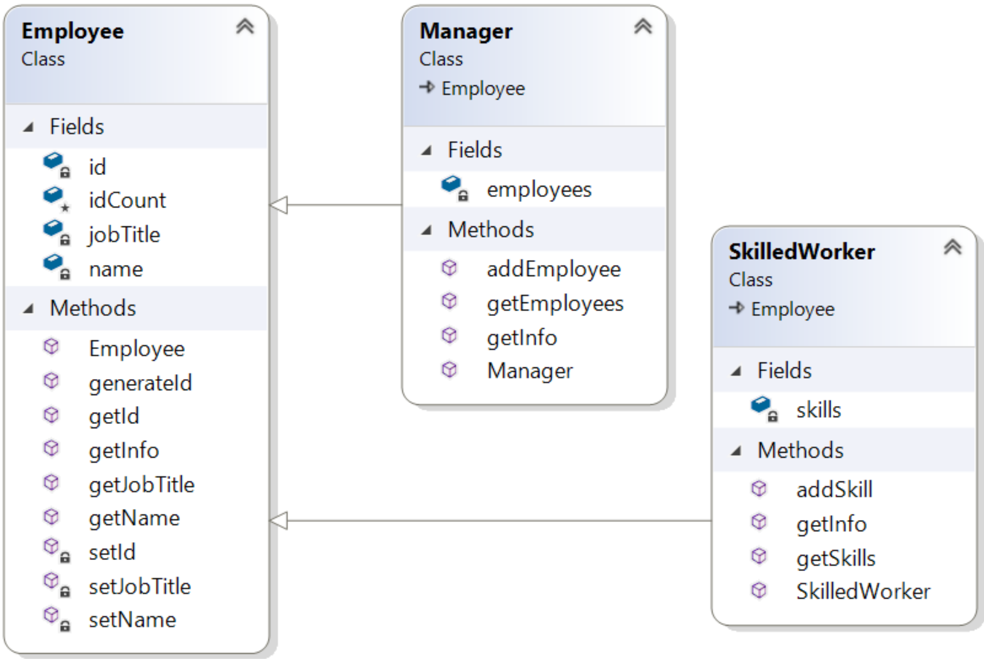
Part 2 — Employee hierarchy

Scenario

In these labs, you will design and create a class hierarchy for an employee tracking system.

Designing the hierarchy

1. Create the following classes. The employee class is provided below. Just copy and paste. We'll need you to concentrate on the other two classes.



2. The Manager class holds an **ArrayList**<Employee> called employees.
3. The **addEmployee(Employee emp)** method adds the **emp** object to the **employees** ArrayList.
4. The **getInfo()** method of the Manager should first gather the manager's details (using super.getInfo()) and then use a for-loop to go through the **employees** ArrayList in order to call their getInfo() method. It should then return the resulting string.

5. The **SkilledWorker** has an **ArrayList** <String> called **skills** which holds the names of skills possessed by a SkilledWorker instance.
6. The **addSkill(String skill)** method adds the **skill** String to **skills**.
7. In the main() method:
    a) create a Manager instance.
    b) create a few regular Employee instances and add them to the manager's employees ArrayList.
    c) create a SkilledWorker object with a few skills and then add the instance to the Manager's employee ArrayList.
    d) call the manager's getInfo() method and display the result.

    Question: Can we add a manager instance to a manager's employees?

To get you started, here is the code for the Employee class:

```
public class Employee {

	private String name;
	private String jobTitle;
	private int id;
	protected static int idCount;

	public String getName() {  return name; }
	private void setName(String name) { this.name = name; }

	public String getJobTitle() { return jobTitle; }
	private void setJobTitle(String jobTitle) { this.jobTitle = jobTitle;}

	private void setId(int id) { this.id = id; }
	public int getId() { return id;  }

	public Employee(String name, String jobTitle ) {
				setId(++Employee.idCount * 10);
		setName(name);
			setJobTitle(jobTitle);
	}

	public String getInfo() {
				String info = "\n**** *****";
				info += "Name: " + getName() + "\n";
				info += "Job Title: "+ getJobTitle() + "\n";
				info += "Employee ID: "+ getId()+ "\n";
					return info;
	}
}
```

QA

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